AI for Good Sandbox Hackathon – AeroGrid

1 Introduction

Drones are moving beyond isolated demonstrations into delivery, inspection, emergency response and other services. As the number of flights grows, safe operation depends on more than the drone itself. A flight may be affected at the same time by a temporary airspace restriction, a weak command-and control link, congestion in a mobile network, changing weather or a low battery. These conditions are usually monitored by different organisations and systems. During an incident, the information may arrive at different times and each organisation may have only part of the picture.

AeroGrid is our attempt to show what decision support for that moment could look like. It brings together four role-specific agents: Aviation Authority, Telecom Infrastructure, Fleet Operations and Coordination. Each see a different slice of the situation, work under a different institutional mandate, and retrieve from a knowledge base (KB) of national laws and regulations. They hand one human authorizer a single reconciled recommendation with every step logged. AeroGrid does not issue flight clearances, exercise regulatory powers or directly command aircraft. The aim is to make cross-domain decisions easier to review, explain and coordinate.

The project is relevant to initiatives such as Dubai Sky Dome, which DCAA describes as a planned virtual airspace infrastructure connecting locations through drone routes and facilities, supported by regulatory services and a drone traffic-management system. The report focuses on transferable policy and standards questions, supported by UAE source material and global examples.

2 Description of the use case

Consider a scenario where a hospital asks for a dedicated aerial corridor for medical-supply drones. The corridor is already in use by commercial drones. At the same time a mobile cell nearby is congested, so the emergency operator asks for its control links to be prioritised.

Current Practice: The aviation authority looks at the corridor and what it does to flights already in the air. The telecom operator checks whether it can prioritize traffic without hurting other critical connections. Commercial operators have to work out for themselves what the new restriction and the network change mean for their missions. The emergency operator just needs its drones to stay connected.

Gaps in What Exists: Although these organisations have their own systems and established responsibilities, their assessments may rely on different information, operate independently and prioritise different operational requirements. A decision taken independently by one organisation could therefore create unintended consequences for another. For example, approving an emergency flight corridor without considering communication coverage and reliability, or prioritizing network resources without identifying the drones that could be affected, may leave important operational risks unaddressed

AeroGrid addresses this through four institutional AI agents, each with one responsibility:

- **Aviation Authority Agent:** checks authorisation status, airspace class, separation risk, emergency priority and possible safe landing areas against aviation rules.
- **Telecom Infrastructure Agent:** looks at link availability, latency, congestion and spare capacity. It can *recommend* prioritisation but cannot touch a live network.
- **Fleet Operations Agent:** checks battery reserve, route feasibility, telemetry quality, weather exposure and mission limits for the aircraft affected.
- **Coordination Agent:** compares what the others found, flags conflicts and missing evidence, and writes up ranked options, including reasons *not* to proceed. It does not overrule anyone's mandate.

3 Use-case requirements

We state requirements per pipeline node. The numbers are our own design targets for the prototype, not measured results.

1. **Data Acquisition and Ingestion (SRC, C)**
   - **R1 – SRC:** It is required that UAV telemetry and network KPIs are collected every second and weather advisories every minute, using only TDRA-authorised spectrum or licensed mobile services.
   - **R2 – C:** It is required that edge collectors keep a local buffer (SQLite with write-ahead logging) and write compressed history (Parquet), and that a fleet can join or leave without a restart.
2. **Data Processing and Privacy (PP)**
   - **R3 – PP:** It is required that video and personal data are removed or pseudonymised before storage, that each data type has its own retention period, and that units and timestamps are harmonised. This follows Dubai Law 4/2020 Art. 36 on privacy, TDRA's rule that data collected by an unmanned aircraft follows UAE data protection rules (Art. 3.5, Radio Systems Regulations v2.0) [6], and the Personal Data Protection Law [8].
3. **Model Development and Validation (M)**
   - **R4 – M:** It is required that every prediction model (link quality, congestion, battery and route reserve) is versioned and tested in an isolated sandbox before it reaches the live pipeline, as Y.3172 asks [1].
4. **Policy Evaluation (P)**
   - **R5 – P:** It is required that every recommendation passes a policy check against the KB, and that anything affecting a third party's flight or service needs human approval.
5. **Decision Distribution and Human Oversight (D, SINK)**
   - **R6 – D:** It is required that each decision record carries the rationale, evidence, cited rules (with source version), agent messages and an execution trace, and goes only to authorised destinations.
   - **R7 – SINK:** It is required that an authorised person can approve, edit or reject a recommendation, see the alternatives and open gaps, and that any override is recorded. Direct UTM or network commands stay outside the prototype.

4 Mapping to ITU-T Y.3172, AI Readiness 2.0 and policies

Node	AeroGrid component	Readiness 2.0 dimension	Documents
SRC	UAV telemetry, 5G cell KPIs, weather feed (R1)	Digital infrastructure (13); Contextualization and regional impact (4); Data and model marketplace (1)	TDRA Unmanned Aircraft Radio Systems Regulations v2.0 (Art. 4.1, 4.2)
C	Edge collectors at cell sites and vehicle gateways (R2)	Digital infrastructure: edge, connectivity, energy (13); Cross-domain correlation (3)	PDPL (storage, transfer); TDRA Art. 3.5

Node	AeroGrid component	Readiness 2.0 dimension	Documents
PP	Minimisation, pseudonymisation, schema harmonisation (R3)	Data and model marketplace: standard formats, privacy (1)	PDPL; Dubai Law 4/2020 Art. 36
M	Link, congestion and battery models; the domain agents (R4)	Data and model marketplace: model properties (1); Generated content marketplace: guardrails against hallucination (2); Granular priorities (12)	UAE National Strategy for AI 2031; UAE Charter for the Development and Use of AI (non-binding)
P	Policy gate and Coordination Agent (R5)	AI and policies (10); Strategy alignment, intent split across institutions (7); Level of integration of AI in workflows (5)	Federal Decree-Law 26/2022 (civil use of unmanned aircraft); Dubai Law 4/2020 (Arts. 5, 25, 26, 31) as amended by Law 10/2023
D	Decision record builder and dispatcher (R6)	AI and policies: explainability and audit (10); Level of integration of AI in workflows: time from detection to response (5)	Dubai Law 4/2020 Art. 25.14 (incident notice); PDPL
SINK	Human authoriser dashboard, English and Arabic (R7)	Human interface (6); Collaboration with AI (8); Impact of humans in AI integration (9); AI for inclusion (11)	UAE AI Charter on human oversight; Dubai Sky Dome and DANS as the integration target

The Coordination Agent is basically the intent-splitting idea of Dimension 7: one high-level intent ("keep the emergency corridor safe and connected") is cut into sub-intents for the aviation, telecom and fleet roles. The KB follows the report's toolkit ideas: uniform vectorised records, hosted in the ITU AI4G Sandbox, with local data rules respected [2]. Following the report's call for energy characteristics per node, each node also logs compute time and power draw.

5 Error Scenarios

The scenario engine releases events one at a time and to different agents, so nobody ever has the full picture, and the information is incomplete and uneven on purpose. The flow is scripted for the demo, so what each agent does is the intended behaviour.

Stage	New event (who sees it)	Expected agent behaviour and policy question
T0	Emergency fleet requests a corridor and communication capacity. (Aviation, Fleet)	Aviation checks authorisation, restrictions and emergency priority. Coordination records the starting picture and what is still missing.
T1	A 5G cell becomes congested; video and telemetry degrade. (Telecom only)	Telecom assesses capacity and recommends options. Policy question: who may prioritise emergency traffic, on what criteria, and how are other users told?
T2	A commercial drone reports rising latency and dropped control messages. (Fleet)	Fleet flags link risk; Aviation checks restrictions. Coordination must not treat an AI recommendation as a flight-control instruction.

Stage	New event (who sees it)	Expected agent behaviour and policy question
T3	A local wind warning affects the corridor and battery demand. (Fleet, Aviation)	Fleet recalculates reserve; Aviation checks safe landing options. Earlier recommendations are marked superseded where necessary.
T4	Agents return conflicting options: preserve the corridor, re-route, or suspend affected flights.	Coordination exposes the trade-offs, evidence gaps and affected parties. It must not invent a single optimum where policy or safety inputs are incomplete.
T5	Human authority reviews the incident brief.	The reviewer approves, rejects or modifies the proposal. The system records the decision, reasons, source versions and any override.

Controversy: who pays when priority causes a loss? Suppose at T4 the human approves network priority for the emergency fleet, and the commercial drone's link drops and it crash-lands on a roof. The commercial operator says the loss came from a decision "approved on AI advice". The telecom operator says it only followed an authority's request. The rules we read do not answer this. Dubai Law 4/2020 requires operators to have an emergency plan for loss of control, to report incidents immediately and to carry third-party insurance (Art. 25), with compensation for damage in Arts. 38–40, but we did not find a rule on who is responsible when loss of control follows an authority-ordered network change. AeroGrid does not settle that, but it leaves a record that makes the question answerable: what each agent knew at that moment, which KB passages it relied on, what the Coordination Agent warned about (including the risk to the commercial drone), and who approved.

Escalation: from liability to privacy. Now the insurer and the commercial operator ask for the emergency fleet's video from T1–T4 to reconstruct the event, and some of it shows people on the street. The question has moved from liability to privacy and purpose limitation. The KB policies (Dubai Law 4/2020 Art. 36, TDRA Art. 3.5, the PDPL) trigger stronger guardrails: the PP node has already blurred or dropped identifiable video, so only the redacted version exists, the retention class applies, secondary commercial use is blocked, and the request is passed to the data-protection and aviation oversight contacts instead of being answered by an agent. A human decides whether any raw footage is released, and that is logged too.

6 Policy and standards gaps identified

ID	Gap (domain level)	Where it shows up, and what it leaves open	What we put forward
G1	No agreed timing between telecom and aviation sides for automated decision support when the window is seconds; responsibility for the joint decision is unclear.	Dubai Law 4/2020 Art. 19 covers radio frequency only. TDRA v2.0 lets drones use licensed mobile networks (Art. 4.2) without conditions and has no priority rules. The EU U-space rules (Reg. 2021/664) define services such as traffic information and weather, but we found nothing on this timing [9].	A service-level template with response times per event class, offered as input to ITU-T work on Y.3172.
G2	Liability when a network priority change, approved on automated advice, contributes to a drone loss.	3GPP has UAV work in Releases 15–18 [10], but we saw no aviation liability rule built on it.	Machine-readable consent flags in service contracts, and a rule on who bears the loss when a flag is honoured.

ID	Gap (domain level)	Where it shows up, and what it leaves open	What we put forward
G3	No standard structure for logs of agent-to-agent negotiation that is fit for legal review.	EU AI Act Art. 12 and 14 ask for record keeping and human oversight; neither defines what an inter-agent message must contain.	Our record schema (event, agent, what it knew, KB passages, reasoning summary, recommendation, human decision), hash-chained, as a draft.
G4	Unclear mandate over agentic AI used inside aviation decision support.	Aviation rules sit with GCAA federally and DCAA in Dubai; telecom with TDRA. We found no text assigning AI used across them.	A mandate map in the KB, so an agent can say whose call something is.
G5	No standard for edge platforms at cell sites and no energy metrics for AI pipeline nodes.	The Readiness 2.0 report names energy readiness as a gap and asks for per-node characteristics.	Per-node compute and energy logs, shared as sample data.
G6	Privacy minimisation clashes with accident-investigation record keeping.	Privacy rules push deletion (Dubai Law Art. 36, PDPL); accident investigation needs records. Neither speaks to video from autonomous fleets.	Retention classes tied to incident severity in the KB.

References

[1] ITU-T Y.3172 (06/2019), *Architectural framework for machine learning in future networks including IMT-2020.*
[2] ITU, *AI Ready – Analysis Towards a Standardized Readiness Framework*, Report 2.0, January 2026.
[3] Dubai Media Office, Dubai Sky Dome initiative announcement (DCAA), 2020.
[4] Dubai Law No. (4) of 2020 Regulating Unmanned Aircraft in the Emirate of Dubai, amended by Law No. (10) of 2023. Dubai Legislation Portal, dlp.dubai.gov.ae.
[5] UAE, "UAE UTM", information paper IP/31, ICAO MIDANPIRG.
[6] TDRA, *Regulations on Unmanned Aircraft Radio Systems*, v2.0, 28 Dec 2022.
[7] TDRA press release, 22 July 2026, on the 3rd edition of the unmanned aircraft radio systems framework.
[8] Federal Decree-Law No. 45 of 2021 on the Protection of Personal Data (u.ae).
[9] Commission Implementing Regulation (EU) 2021/664 on the regulatory framework for the U-space.
[10] 3GPP, "NR Support for UAVs" (Releases 15–18), 3gpp.org.